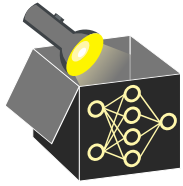
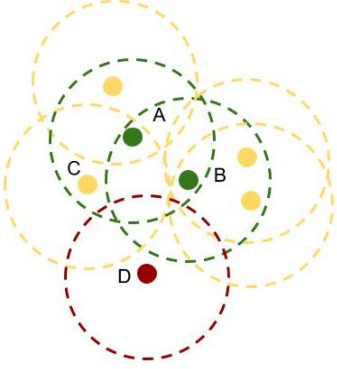
# Interpretable Machine Learning



## Counterfactuals and Adversarial Examples
## Increasing Trust in Explanations

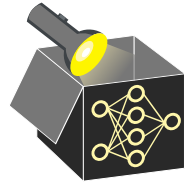



**Learning goals**

- Understand the aspects that undermine users' trust in an explanation
- Learn diagnostic tools that could increase trust

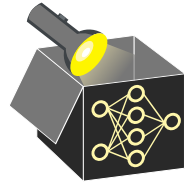
# MOTIVATION & IMPORTANT PROPERTIES

- Local explanations should not only make a model interpretable but also reveal if the model is trustworthy

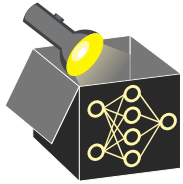


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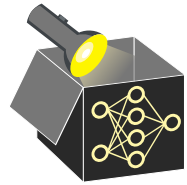


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     - Failure case: generation is based on inputs in areas where the model was trained with little or no training data (extrapolation)

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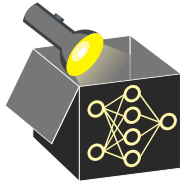


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     ⇝ measure how robust an IML method is to small changes in the input data or parameters

     ⇝ Is an observation out-of-distribution?

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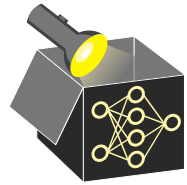
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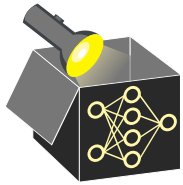
     $\rightsquigarrow$ Is an observation out-of-distribution?
- Failing in one of these $\rightsquigarrow$ undermining users’ trust in the explanations
  $\rightsquigarrow$ undermining trust in the model

# OUT-OF-DISTRIBUTION (OOD) DETECTION

- Models are unreliable in areas with little data support
  $\rightsquigarrow$ explanations from local explanation methods are unreliable

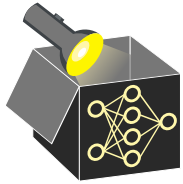


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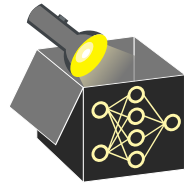


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- Two very simple and intuitive approaches
  - Classifier for out-of-distribution
  - Clustering
- More complicated also possible, e.g., variational autoencoders
  - "Daxberger et al." 2020

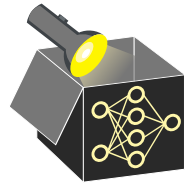
# OOD DETECTION: OOD-CLASSIFIER

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- Idea: Hallucinate new (ood) data by randomly sampling data points

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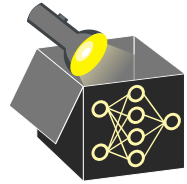
⇝ Learn a binary classifier to distinguish between the origins of the data

- Study whether an explanation approach can be fooled ▸ "Dylan Slack et al." 2020
  - Hide bias in the true (deployed) model, but use an unbiased model for all out-of-distribution samples

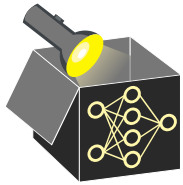
⇝ Important way to diagnose an explanation approach

# OOD DETECTION: CLUSTERING VIA DBSCAN

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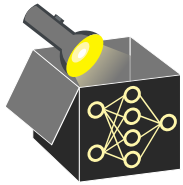


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- For this method, we define an $\epsilon$-neighborhood:
  Given a dataset $X = \{\boldsymbol{x}^{(i)}\}_{i=1}^{n}$, an $\epsilon$-neighborhood for $\boldsymbol{x} \in \mathcal{X}$ is defined as

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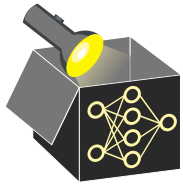


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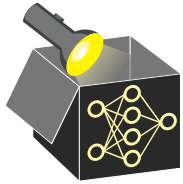


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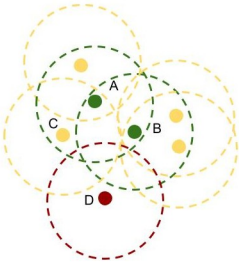


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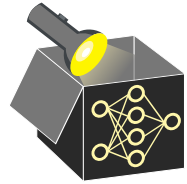
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- Noise points
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  - Not part of any cluster

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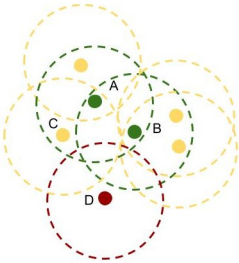



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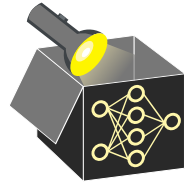


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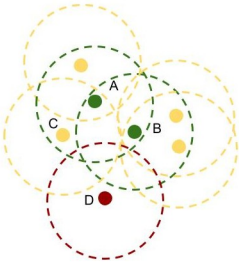



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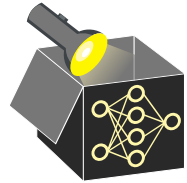


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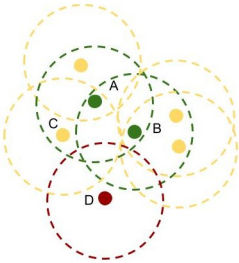



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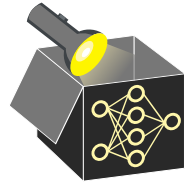


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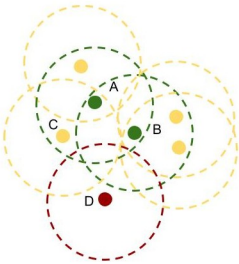



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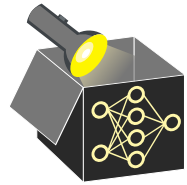


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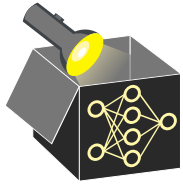
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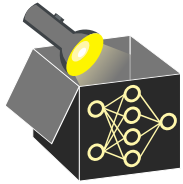
- Disadvantages:
  - Depending on the distance metric $d(\cdot)$, DBSCAN could suffer from the "curse of dimensionality"
  - The choice of $\epsilon$ and $m$ is not clear a-priori

# ROBUSTNESS

- Differentiate between different kinds of uncertainty:
  1. **Explanation uncertainty**: Explanation changes if we repeat the process, e.g., the explanation could differ depending on which subset of data we use for the expl. method and which hyperparams

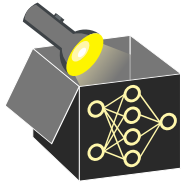


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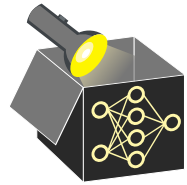
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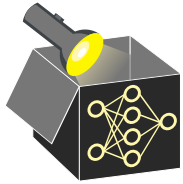
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- We focus on explanation uncertainty
  - Even with the same model and same (or similar) data points, we can receive different explanations

# ROBUSTNESS MEASURE FOR LIME AND SHAP

- Objective: Similar explanations for similar inputs (in a neighborhood)



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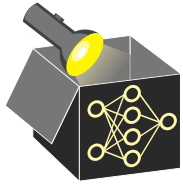


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- For LIME and SHAP, notion of stability based on **locally Lipschitz continuity**
  ▸ "Alvarez-Melis and Jaakkola" 2018 :

  An explanation method $g : \mathcal{X} \rightarrow \mathbb{R}^m$ is locally Lipschitz if
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  Note that, for LIME, $g$ returns the $m$ coefficients of the surrogate model

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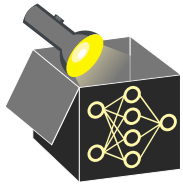
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  Note that, for LIME, $g$ returns the $m$ coefficients of the surrogate model
- According to this, we can quantify the robustness of explanation models in terms of $\omega$:

  $\rightsquigarrow$ The closer $\omega$ is to 0, the more robust our explanation method is

# ROBUSTNESS MEASURE FOR LIME AND SHAP



- Objective: Similar explanations for similar inputs (in a neighborhood)
- For LIME and SHAP, notion of stability based on **locally Lipschitz continuity** ▸ "Alvarez-Melis and Jaakkola" 2018:

  An explanation method $g : \mathcal{X} \to \mathbb{R}^m$ is locally Lipschitz if
  - for every $\boldsymbol{x}_0 \in \mathcal{X}$ there exist $\delta > 0$ and $\omega \in \mathbb{R}$
  - such that $||\boldsymbol{x} - \boldsymbol{x}_0|| < \delta$ implies $||g(\boldsymbol{x}) - g(\boldsymbol{x}_0)|| < \omega||\boldsymbol{x} - \boldsymbol{x}_0||$

  Note that, for LIME, $g$ returns the $m$ coefficients of the surrogate model
- According to this, we can quantify the robustness of explanation models in terms of $\omega$:

  $\rightsquigarrow$ The closer $\omega$ is to 0, the more robust our explanation method is
- $\omega$ is rarely known a-priori but it could be estimated as follows:

$$\hat{\omega}_X(\boldsymbol{x}) \in \underset{\boldsymbol{x}^{(i)} \in \mathcal{N}_\epsilon(\boldsymbol{x})}{\arg\max} \frac{||g(\boldsymbol{x}) - g(\boldsymbol{x}^{(i)})||_2}{d(\boldsymbol{x}, \boldsymbol{x}^{(i)})}$$

where $\mathcal{N}_\epsilon(\boldsymbol{x})$ is the $\epsilon$-neighborhood of $\boldsymbol{x}$